Activity 05 · Segments and Customer Value

TGS-2022017520 · v14.0 · Synthetic training data

Goal

A fictional programme has working professionals and career switchers. Behavioural segments should guide research questions and consented messaging, with customer value calculated using explicit margin and retention assumptions.

Before you start

Use a copied activity folder,Claude Cowork or Claude Code as specified,and a spreadsheet or Python3 for independent checks. No production credentials are needed. Product access varies;CSV fallback remains valid.

Mock data and assets

mock-data/segments.csv

All named customers,competitors,campaigns and values are fictional. Assets include brand,source,measurement and decision contracts.

Mechanism

Choose observable non-sensitive variables, compute group sizes and values, validate distinct needs with evidence, avoid assigning personas beyond available data.

Digital Marketing’s consumer-behaviour and relationship framework is adapted into observed segment value plus testable needs. Generated personas are hypotheses only; no demographic identity is inferred from behaviour.

Worked calculation

Working margin/customer = (24000-12000)/40 = $300. Simple one-period retained margin estimate = $300 × 0.7 = $210; not lifetime value.

Step-by-step

1. Inspect segment sizes and reconcile total to 80 customers.
2. Read assets/persona-template.md and identify observation versus hypothesis fields.
3. Ask Cowork Prompt 1 for a segment-value table.
4. Calculate contribution margin independently for each row.
5. Confirm $300,$200,$500 unit margins and distinguish total value from unit value.
6. Draft two validation questions per segment; avoid leading questions.
7. Apply Prompt 2 to remove unsupported demographic assumptions.
8. Save the table and persona hypotheses under outputs.

Prompt1

Analyse synthetic segments.csv. Calculate customers and contribution margin per customer. Create evidence-constrained persona hypotheses using only observed fields. Do not invent age, gender or income. Label retention assumptions and propose validation questions.

Prompt2

Challenge the previous result. Find denominator errors,unsupported claims,bias,privacy risks and unexecuted tool boundaries. Cite exact row IDs or fields. Revise the report without inventing data.

Acceptance evidence

Margins/customer 300,200,500; total customers 80; no invented demographic traits; retained-margin assumptions explicit.

Failure and recovery

The agent invents an age range and income for each segment. Neither field exists; the persona must remain evidence-constrained.

Use behavioural attributes and minimum group size for reporting; label retention as an assumption; suppress identifying small-group combinations.

Submit

segment-value.csv and persona-hypotheses.md

Save outputs in this working copy. Keep an execution note stating which product actions actually ran and which used fallback.